

Independant Component Analysis (ICA) and its non linear extension through VAEs

Objectives

The objectives of this project was first to define **ICA** [1], and to give some rigorous justification of its ideas. Then to implement it on different datasets, to highlight some technical aspect. The second part of the project was to understand the link with **identifiability theory of VAEs** [2] and to implement it.

Mathematical foundation of ICA

Let $A \in \mathcal{M}_n(\mathbb{R})$ and $T > 0$. The aim of linear ICA is to reconstruct a signal $s = (s_1, \dots, s_n)^T: [0, T] \rightarrow \mathbb{R}$, from the knowledge $x := As$.

The huge idea of ICA, is that up to a notion of **non-gaussianity**, the $\pm s_i$ will be the local maximum for this notion of the function $\phi(w) := w^T x$.

The principal notion of non-gaussianity is the **negentropy**, which is motivated by information theory.

ICA : Optimisation & Decorrelation

Deflation ICA $y_i := w_i^T X$

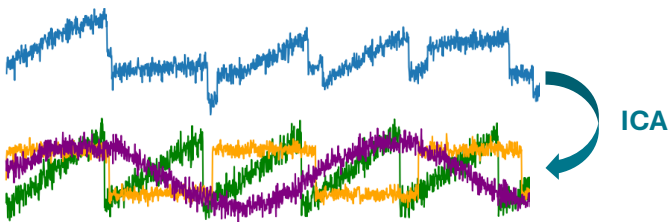
$$w_i = \mathbb{E}\{X \cdot g(y_i)^T\} - \mathbb{E}\{g'(y_i)\} \cdot w$$
$$w_i = w_i - \sum_{j=1}^{i-1} (w_i^T w_j) w_j$$

Symmetric ICA

$$W = W + \text{diag}(\alpha)(\text{diag}(\beta) + \mathbb{E}\{g(y)y^T\}W)$$
$$W = (WW^T)^{-1/2}W$$
$$\beta_i := -\mathbb{E}\{y_i g(y_i)\} \quad \alpha_i := -1/(\beta_i + \mathbb{E}\{g'(y_i)\})$$

ICA Experiments : Source Separation

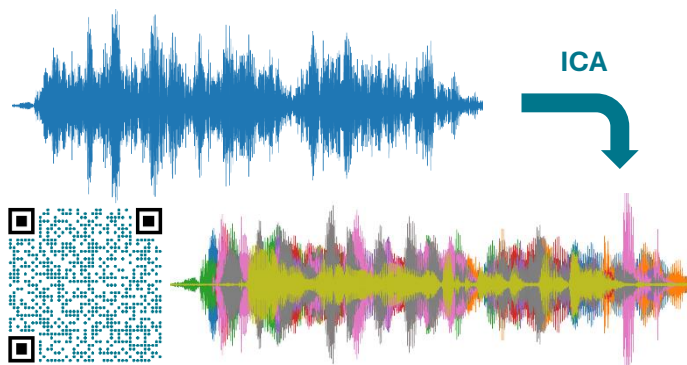
Synthetic data



Open-source data



Recorded audios



Identifiability of VAEs and non-linear ICA

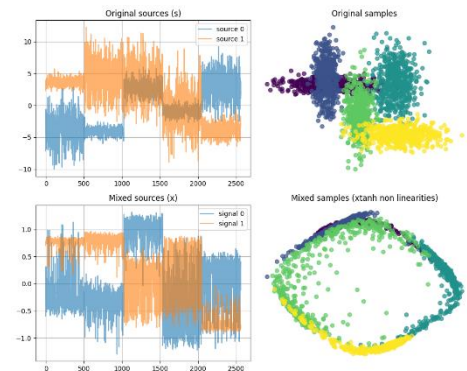
The identifiability is the objective of ICA. It is quite a similar problem in VAEs, because there are few possible prior, which will give the same reconstruction.

We consider $x = f(x) + \epsilon$, f injective and ϵ random. It is very similar to ICA, but now f can be **non-linear**. To reconstruct f and z , we will see f as the parameter of a neural network in the VAE framework

$$p(x, z | u) = p_f(x | z) \prod_i p_{T_i, \lambda_i}(z_i, u)$$

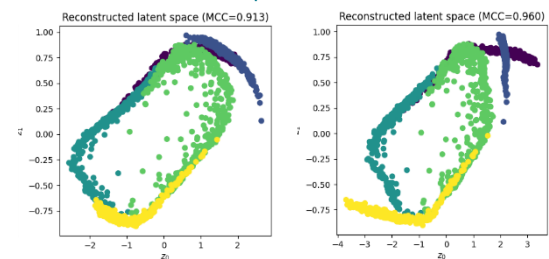
Implementation

We implemented **non-linear ICA** using **Identifiable Variational Autoencoders** (iVAEs) as described in [2]. We generated independent non-stationary Gaussian sources s conditioned by their segment indices u , and the associated observed signals x as non-linear combinations of the sources using a Multi-Layer Perceptron (MLP).

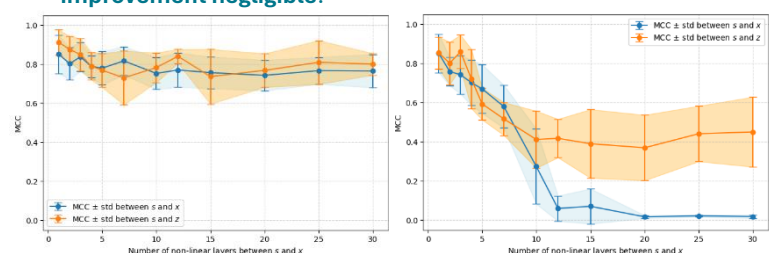


Results

Although we achieved **high MCC scores** (> 0.9), we found the metric deceptive as it often masked **poor geometric reconstruction** in the latent space.



Furthermore, the iVAE only significantly outperforms the raw signal in **complex mixing regimes** (e.g., deep Leaky-ReLU networks). In simpler scenarios, the signal itself retains high correlation with the sources, rendering the **model's improvement negligible**.



References

- [1] E. Oja A. Hyvärinen. Independent component analysis: algorithms and applications.
- [2] R. Pio Monti A. Hyvärinen I. Khemakhem, D. P. Kingma. Variational autoencoders and nonlinear ica: A unifying framework.
- [3] Lindaswala Muda, Mumtaj Begam, and Irravain Elamvazuthi. Voice recognition algorithms using mel frequency cepstral coefficient (mfcc) and dynamic time warping (dtw) techniques.